

2024

B.Tech. 3rd Semester End-Term Examination

Civil Engineering

FLUID MECHANICS

Full Marks – 70

Time – Three hours

The figures in the margin indicate full marks
for the questions.

Answer question No. 1 and any *four* from the rest.

1. Answer the following MCQ: 1 × 10
- (i) If 5m^3 of a certain oil weighs 40 KN, what is specific weight and mass density respectively? ($g = 10 \text{ m/sec}^2$)
- (a) 8000 N/m^3 and 800 N/m^3
 - (b) 8000 N/m^3 and 800 kg/m^3
 - (c) 8000 kg/m^3 and 800 kg/m^3
 - (d) 800 N/m^3 and 8000 kg/m^3
- (ii) Compressibility is the reciprocal of
- (a) Bulk modulus of elasticity
 - (b) Shear modulus of elasticity
 - (c) Young's modulus of elasticity
 - (d) None of the above
- (iii) When a body is immersed in a fluid, the upward force exerted by the fluid is equal to the
- (a) Weight of the body
 - (b) Volume of water displaced by of the body
 - (c) Weight of the fluid displaced by the body
 - (d) None of the above

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- (iv) The distance of Centre of pressure from free surface of liquid is independent of
- The density of liquid
 - The moment of inertia
 - The surface area
 - The distance of C. G. from free surface of liquid
- (v) A pitot tube is used for measuring
- Velocity of flow
 - Pressure of flow
 - Flow rate
 - Total energy
- (vi) The concept of stream function which is based on the principle of continuity is applicable to
- Irrotational flow only
 - Two- dimensional flow only
 - Three- dimensional flow
 - Uniform flow only
- (vii) The horizontal to vertical side slope in case of Cippoletti weir is
- 1:1
 - 1:2
 - 1:3
 - 1:4
- (viii) Francis discovered that each end contraction reduces the crest length by
- 0.01H
 - 1.01H
 - 1.1H
 - 0.1H
- (ix) The energy loss in a pipe line is due to
- Surface roughness only
 - Viscous action only
 - Friction offered by pipe wall as well as by viscous action
 - None of the above
- (x) In which of the following methods of dimensional analysis, a functional relationship of some variables is expressed in the form of an exponential equation, which must be dimensionally homogeneous?
- Buckingham's π -method
 - Rayleigh's method
 - Bridgman's method
 - Matrix-tensor method

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2. (a) The surface tension of water in contact with air at 20°C is given by 0.0716 N/m . The pressure inside the droplet of water is 0.0147 N/cm^2 and is excess of outside pressure. Calculate the diameter of the droplet of water. 3
- (b) State Newton's law of viscosity. Give the classification of fluids with examples based on above law. 4
- (c) Define the terms: 8
- (i) Stream line and streak line
- (ii) Hydraulic gradient line and Total energy line
- (iii) Forced vortex flow and Free vortex flow
- (iv) Orifice and Mouthpiece
3. (a) A circular plate of diameter 1 m is immersed in water with its plane making an angle of 45° with the horizontal. The centre of the plate is at a depth of 2 m below the free surface. Calculate the total force on one side of the plate and the location of the centre of pressure. 3
- (b) "A floating body is in stable equilibrium if the metacentre is above the centre of gravity of the body". Explain the statement with a neat sketch. 3
- (c) A vertical U- tube of uniform cross- section contains mercury in both its arms. A water column of length 10 cm is introduced into one of the arms. Oil of relative density 0.7 is poured in the other arm until the upper surface of the oil and water are in the same horizontal level. Find the length of the oil column. 4
- (d) A solid cylinder of 3 m diameter has a length of 6 m . It is made up of a material whose specific gravity is 0.8 and is floating in water with its axis vertical. Find its metacentric height and state whether its equilibrium is stable or unstable. 5
4. (a) The stream function for a two-dimensional flow is given by $\psi = 2xy$, calculate the velocity at the point $P (2, 3)$. Find the velocity potential function ϕ . 7
- (b) How is vena-contracta defined? Derive an expression for discharge through partially submerged orifice. 2 + 6
5. (a) Explain how you will find the velocity at a point in a liquid flowing in an open channel using a pitot tube. 3
- (b) A horizontal venturimeter whose inlet and throat diameters are 30 cm and 15 cm respectively is used to measure the flow of water. The differential manometer connected to the inlet and throat shows a reading of 18 cm of mercury. Find the rate of flow if $C_d = 0.97$. 4
- (c) State and prove Bernoulli's theorem. 8

6. (a) Derive Darcy-Weisbach formula for calculating loss of head due to friction in pipe. 5
- (b) The discharge through a pipe is 200 litres/s. Find the loss of head when the pipe is suddenly enlarged from 150 mm to 300mm diameter. 3
- (c) Find the discharge through a trapezoidal notch which is 1.2m wide at the top and 0.5 m at the bottom and is 40cm in height. The head of water on the notch is 30cm. Assume C_d for rectangular portion as 0.62 while for triangular portion as 0.6. 7
7. (a) What is a dimensionally homogeneous equation? Explain with an example. 3
- (b) An open tank 4 m long, 2 m wide and 1.5 m deep is filled with an oil of specific gravity 0.8 upto a depth of 1 m. Determine the maximum horizontal acceleration, which can be given to the tank, so that the oil does not spill out. 4
- (c) Show that the resistance R to the motion of a sphere of diameter D moving with uniform velocity V through a fluid of density ρ and viscosity μ is given by 8

$$R = \frac{\mu^2}{\rho} f\left(\frac{DV\rho}{\mu}\right)$$

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